

is the temperature on the i th day of the month. Suppose that $E[T_i] = 80$ for all i , and that T_i and T_j have covariance

$$\text{Cov}[T_i, T_j] = \frac{36}{1 + |i - j|} \quad (8.67)$$

Define the daily average temperature as

$$Y = \frac{T_1 + T_2 + \cdots + T_{31}}{31}. \quad (8.68)$$

Based on this model, write a MATLAB program `p=julytemps(T)` that calculates $P[Y \geq T]$, the probability that the daily average temperature is at least T degrees.

Further Reading: [WS01] and [PP02] make extensive use of vectors and matrices. To go deeply into vector random variables, students can use [Str98] to gain a firm grasp of principles of linear algebra.

Problems

Difficulty: ● Easy ■ Moderate ♦ Difficult ♦♦ Experts Only

8.1.1 ● For random variables $X_1, \dots, X_n$ in Problem 5.10.3, let $\mathbf{X} = [X_1 \ \cdots \ X_n]'$. What is $f_{\mathbf{X}}(\mathbf{x})$?

8.1.2 ● Random vector $\mathbf{X}$ has PDF

$$f_{\mathbf{X}}(\mathbf{x}) = \begin{cases} c\mathbf{a}'\mathbf{x} & \mathbf{0} \leq \mathbf{x} \leq \mathbf{1}, \\ 0 & \text{otherwise,} \end{cases}$$

where $\mathbf{a} = [a_1 \ \cdots \ a_n]'$ is a vector with each component $a_i > 0$. What is c ?

8.1.3 ■ Given $f_{\mathbf{X}}(\mathbf{x})$ with $c = 2/3$ and $a_1 = a_2 = a_3 = 1$ in Problem 8.1.2, find the marginal PDF $f_{X_3}(x_3)$.

8.1.4 ■ $\mathbf{X} = [X_1 \ X_2 \ X_3]'$ has PDF

$$f_{\mathbf{X}}(\mathbf{x}) = \begin{cases} 6 & 0 \leq x_1 \leq x_2 \leq x_3 \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Let $\mathbf{U} = [X_1 \ X_2]'$, $\mathbf{V} = [X_1 \ X_3]'$ and $\mathbf{W} = [X_2 \ X_3]'$. Find the marginal PDFs $f_{\mathbf{U}}(\mathbf{u})$, $f_{\mathbf{V}}(\mathbf{v})$ and $f_{\mathbf{W}}(\mathbf{w})$.

8.1.5 ■ A wireless data terminal has three messages waiting for transmission. After sending a message, it expects an acknowl-

edgement from the receiver. When it receives the acknowledgement, it transmits the next message. If the acknowledgement does not arrive, it sends the message again. The probability of successful transmission of a message is p independent of other transmissions. Let $\mathbf{K} = [K_1 \ K_2 \ K_3]'$ be the three-dimensional random vector in which K_i is the total number of transmissions when message i is received successfully. (K_3 is the total number of transmissions used to send all three messages.) Show that

$$P_{\mathbf{K}}(\mathbf{k}) = \begin{cases} p^3(1-p)^{k_3-3} & k_1 < k_2 < k_3; \\ & k_i \in \{1, 2, \dots\}, \\ 0 & \text{otherwise.} \end{cases}$$

8.1.6 ■ From the joint PMF $P_{\mathbf{K}}(\mathbf{k})$ in Problem 8.1.5, find the marginal PMFs

- (a) $P_{K_1, K_2}(k_1, k_2)$,
- (b) $P_{K_1, K_3}(k_1, k_3)$,
- (c) $P_{K_2, K_3}(k_2, k_3)$,
- (d) $P_{K_1}(k_1)$, $P_{K_2}(k_2)$, and $P_{K_3}(k_3)$.

8.1.7 ■ Let $\mathbf{N}$ be the r -dimensional random vector with the multinomial PMF given in

Example 5.21 with $n > r > 2$:

$$P_{\mathbf{N}}(\mathbf{n}) = \binom{n}{n_1, \dots, n_r} p_1^{n_1} \cdots p_r^{n_r}.$$

- (a) What is the joint PMF of N_1 and N_2 ? Hint: Consider a new classification scheme with categories: s_1 , s_2 , and “other.”
- (b) Let $T_i = N_1 + \cdots + N_i$. What is the PMF of T_i ?
- (c) What is the joint PMF of T_1 and T_2 ?

8.1.8 ■ The random variables $Y_1, \dots, Y_4$ have the joint PDF

$$f_{\mathbf{Y}}(\mathbf{y}) = \begin{cases} 24 & 0 \leq y_1 \leq y_2 \leq y_3 \leq y_4 \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Find the marginal PDFs $f_{Y_1, Y_4}(y_1, y_4)$, $f_{Y_1, Y_2}(y_1, y_2)$, and $f_{Y_1}(y_1)$.

8.1.9 ♦ As a generalization of the message transmission system in Problem 8.1.5, consider a terminal that has n messages to transmit. The components k_i of the n -dimensional random vector $\mathbf{K}$ are the total number of messages transmitted when message i is received successfully.

- (a) Find the PMF of $\mathbf{K}$.
- (b) For each $j \in \{1, 2, \dots, n-1\}$, find the marginal PMF $P_{K_1, \dots, K_j}(k_1, \dots, k_j)$.
- (c) For each $i \in \{1, 2, \dots, n\}$, find the marginal PMF $P_{K_i}(k_i)$.

Hint: These PMFs are members of a family of discrete random variables in Appendix A.

8.2.1 ● The n components X_i of random vector $\mathbf{X}$ have $E[X_i] = 0$, $\text{Var}[X_i] = \sigma^2$. What is the covariance matrix $\mathbf{C}_{\mathbf{X}}$?

8.2.2 ● The 4-dimensional random vector $\mathbf{X}$ has PDF

$$f_{\mathbf{X}}(\mathbf{x}) = \begin{cases} 1 & 0 \leq x_i \leq 1, i = 1, 2, 3, 4, \\ 0 & \text{otherwise.} \end{cases}$$

Are the four components of $\mathbf{X}$ independent random variables?

8.2.3 ● As in Example 8.1, the random vector $\mathbf{X}$ has PDF

$$f_{\mathbf{X}}(\mathbf{x}) = \begin{cases} 6e^{-\mathbf{a}'\mathbf{x}} & \mathbf{x} \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

where $\mathbf{a} = [1 \ 2 \ 3]'$. Are the components of $\mathbf{X}$ independent random variables?

8.2.4 ● The PDF of the 3-dimensional random vector $\mathbf{X}$ is

$$f_{\mathbf{X}}(\mathbf{x}) = \begin{cases} e^{-x_3} & 0 \leq x_1 \leq x_2 \leq x_3, \\ 0 & \text{otherwise.} \end{cases}$$

Are the components of $\mathbf{X}$ independent random variables?

8.2.5 ■ The random vector $\mathbf{X}$ has PDF

$$f_{\mathbf{X}}(\mathbf{x}) = \begin{cases} e^{-x_3} & 0 \leq x_1 \leq x_2 \leq x_3, \\ 0 & \text{otherwise.} \end{cases}$$

Find the marginal PDFs $f_{X_1}(x_1)$, $f_{X_2}(x_2)$, and $f_{X_3}(x_3)$.

8.3.1 ● Discrete random vector $\mathbf{X}$ has PMF $P_{\mathbf{X}}(\mathbf{x})$. Prove that for an invertible matrix $\mathbf{A}$, $\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{b}$ has PMF

$$P_{\mathbf{Y}}(\mathbf{y}) = P_{\mathbf{X}}(\mathbf{A}^{-1}(\mathbf{y} - \mathbf{b})).$$

8.3.2 ■ In the message transmission problem, Problem 8.1.5, the PMF for the number of transmissions when message i is received successfully is

$$P_{\mathbf{K}}(\mathbf{k}) = \begin{cases} p^3(1-p)^{k_3-3} & k_1 < k_2 < k_3; \\ & k_i \in \{1, 2, \dots\}, \\ 0 & \text{otherwise.} \end{cases}$$

Let $J_3 = K_3 - K_2$, the number of transmissions of message 3; $J_2 = K_2 - K_1$, the number of transmissions of message 2; and $J_1 = K_1$, the number of transmissions of message 1. Derive a formula for $P_{\mathbf{J}}(\mathbf{j})$, the PMF of the number of transmissions of individual messages.

8.3.3 ■ In an automatic geolocation system, a dispatcher sends a message to six trucks in a fleet asking their locations. The waiting times for responses from the six trucks

are iid exponential random variables, each with expected value 2 seconds.

- What is the probability that all six responses will arrive within 5 seconds?
- If the system has to locate all six vehicles within 3 seconds, it has to reduce the expected response time of each vehicle. What is the maximum expected response time that will produce a location time for all six vehicles of 3 seconds or less with probability of at least 0.9?

8.3.4♦♦ Let $X_1, \dots, X_n$ denote n iid random variables with PDF $f_X(x)$ and CDF $F_X(x)$. What is the probability $P[X_n = \max\{X_1, \dots, X_n\}]$?

8.4.1● Random variables X_1 and X_2 have zero expected value and variances $\text{Var}[X_1] = 4$ and $\text{Var}[X_2] = 9$. Their covariance is $\text{Cov}[X_1, X_2] = 3$.

- Find the covariance matrix of $\mathbf{X} = [X_1 \ X_2]'$.
- Find the covariance matrix of $\mathbf{Y} = [Y_1 \ Y_2]'$ given by

$$\begin{aligned} Y_1 &= X_1 - 2X_2, \\ Y_2 &= 3X_1 + 4X_2. \end{aligned}$$

8.4.2● Let $X_1, \dots, X_n$ be iid random variables with expected value 0, variance 1, and covariance $\text{Cov}[X_i, X_j] = \rho$, for $i \neq j$. Use Theorem 8.8 to find the expected value and variance of the sum $Y = X_1 + \dots + X_n$.

8.4.3● The two-dimensional random vector $\mathbf{X}$ and the three-dimensional random vector $\mathbf{Y}$ are independent and $E[\mathbf{Y}] = 0$. What is the vector cross-correlation $\mathbf{R}_{\mathbf{X}\mathbf{Y}}$?

8.4.4● The four-dimensional random vector $\mathbf{X}$ has PDF

$$f_{\mathbf{X}}(\mathbf{x}) = \begin{cases} 1 & 0 \leq x_i \leq 1, i = 1, 2, 3, 4 \\ 0 & \text{otherwise.} \end{cases}$$

Find the expected value vector $E[\mathbf{X}]$, the correlation matrix $\mathbf{R}_{\mathbf{X}}$, and the covariance matrix $\mathbf{C}_{\mathbf{X}}$.

8.4.5● The random vector $\mathbf{Y} = [Y_1 \ Y_2]'$ has covariance matrix $\mathbf{C}_{\mathbf{Y}} = \begin{bmatrix} 25 & \gamma \\ \gamma & 4 \end{bmatrix}$ where γ is a constant. In terms of γ , what is the correlation coefficient ρ_{Y_1, Y_2} of Y_1 and Y_2 ? For what values of γ is $\mathbf{C}_{\mathbf{Y}}$ a valid covariance matrix?

8.4.6● In the message transmission system in Problem 8.1.5, the solution to Problem 8.3.2 is a formula for the PMF of $\mathbf{J}$, the number of transmissions of individual messages. For $p = 0.8$, find the expected value vector $E[\mathbf{J}]$, the correlation matrix $\mathbf{R}_{\mathbf{J}}$, and the covariance matrix $\mathbf{C}_{\mathbf{J}}$.

8.4.7■ In the message transmission system in Problem 8.1.5,

$$P_{\mathbf{K}}(\mathbf{k}) = \begin{cases} p^3(1-p)^{k_3-3}; & k_1 < k_2 < k_3; \\ & k_i \in \{1, 2, \dots\}, \\ 0 & \text{otherwise.} \end{cases}$$

For $p = 0.8$, find the expected value vector $E[\mathbf{K}]$, the covariance matrix $\mathbf{C}_{\mathbf{K}}$, and the correlation matrix $\mathbf{R}_{\mathbf{K}}$.

8.4.8■ Random vector $\mathbf{X} = [X_1 \ X_2]'$ has PDF

$$f_{\mathbf{X}}(\mathbf{x}) = \begin{cases} 10e^{-5x_1-2x_2} & x_1 \geq 0, x_2 \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

- Find $f_{X_1}(x_1)$ and $f_{X_2}(x_2)$.
- Derive the expected value vector $\boldsymbol{\mu}_{\mathbf{X}}$ and covariance matrix $\mathbf{C}_{\mathbf{X}}$.
- Let $\mathbf{Z} = \mathbf{A}\mathbf{X}$, where $\mathbf{A} = \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}$. Find the covariance matrix of $\mathbf{Z}$.

8.4.9■ As in Quiz 5.10 and Example 5.23, the 4-dimensional random vector $\mathbf{Y}$ has PDF

$$f_{\mathbf{Y}}(\mathbf{y}) = \begin{cases} 4 & 0 \leq y_1 \leq y_2 \leq 1; \\ & 0 \leq y_3 \leq y_4 \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Find the expected value vector $E[\mathbf{Y}]$, the correlation matrix $\mathbf{R}_{\mathbf{Y}}$, and the covariance matrix $\mathbf{C}_{\mathbf{Y}}$.

8.4.10 ■ $\mathbf{X} = [X_1 \ X_2]'$ is a random vector with $E[\mathbf{X}] = [0 \ 0]'$ and covariance matrix

$$\mathbf{C}_X = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}.$$

For some ω satisfying $0 \leq \omega \leq 1$, let $Y = \sqrt{\omega}X_1 + \sqrt{1-\omega}X_2$. What value (or values) of ω will maximize $E[Y^2]$?

8.4.11 ■ The two-dimensional random vector $\mathbf{Y}$ has PDF

$$f_Y(\mathbf{y}) = \begin{cases} 2 & \mathbf{y} \geq \mathbf{0}, [1 \ 1] \mathbf{y} \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Find the expected value vector $E[\mathbf{Y}]$, the correlation matrix $\mathbf{R}_Y$, and the covariance matrix $\mathbf{C}_Y$.

8.4.12 ♦ Let $\mathbf{X}$ be a random vector with correlation matrix $\mathbf{R}_X$ and covariance matrix $\mathbf{C}_X$. Show that $\mathbf{R}_X$ and $\mathbf{C}_X$ are both positive semidefinite by showing that for any nonzero vector $\mathbf{a}$,

$$\begin{aligned} \mathbf{a}'\mathbf{R}_X\mathbf{a} &\geq 0, \\ \mathbf{a}'\mathbf{C}_X\mathbf{a} &\geq 0. \end{aligned}$$

8.5.1 ● $\mathbf{X}$ is the 3-dimensional Gaussian random vector with expected value $\boldsymbol{\mu}_X = [4 \ 8 \ 6]'$ and covariance

$$\mathbf{C}_X = \begin{bmatrix} 4 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 4 \end{bmatrix}.$$

Calculate

- the correlation matrix, $\mathbf{R}_X$,
- the PDF of the first two components of $\mathbf{X}$, $f_{X_1, X_2}(x_1, x_2)$,
- the probability that $X_1 > 8$.

8.5.2 ● $\mathbf{X} = [X_1 \ X_2]'$ is the Gaussian random vector with $E[\mathbf{X}] = [0 \ 0]'$ and covariance matrix

$$\mathbf{C}_X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

What is the PDF of $Y = [2 \ 1] \mathbf{X}$?

8.5.3 ● Given the Gaussian random vector $\mathbf{X}$ in Problem 8.5.1, $\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{b}$, where

$$\mathbf{A} = \begin{bmatrix} 1 & 1/2 & 2/3 \\ 1 & -1/2 & 2/3 \end{bmatrix}$$

and $\mathbf{b} = [-4 \ -4]'$. Calculate

- the expected value $\boldsymbol{\mu}_Y$,
- the covariance $\mathbf{C}_Y$,
- the correlation $\mathbf{R}_Y$,
- the probability that $-1 \leq Y_2 \leq 1$.

8.5.4 ● Let $\mathbf{X}$ be a Gaussian ($\boldsymbol{\mu}_X, \mathbf{C}_X$) random vector. Given a vector $\mathbf{a}$, find the expected value and variance of $Y = \mathbf{a}'\mathbf{X}$. Is Y a Gaussian random variable?

8.5.5 ■ Random variables X_1 and X_2 have zero expected value. The random vector $\mathbf{X} = [X_1 \ X_2]'$ has a covariance matrix of the form

$$\mathbf{C} = \begin{bmatrix} 1 & \alpha \\ \beta & 4 \end{bmatrix}.$$

- For what values of α and β is $\mathbf{C}$ a valid covariance matrix?
- For what values of α and β can $\mathbf{X}$ be a Gaussian random vector?
- Suppose now that α and β satisfy the conditions in part (b) and $\mathbf{X}$ is a Gaussian random vector. What is the PDF of X_2 ? What is the PDF of $W = 2X_1 - X_2$?

8.5.6 ■ The Gaussian random vector $\mathbf{X} = [X_1 \ X_2]'$ has expected value $E[\mathbf{X}] = \mathbf{0}$ and covariance matrix $\mathbf{C}_X = \begin{bmatrix} \sigma_1^2 & 1 \\ 1 & \sigma_2^2 \end{bmatrix}$.

- Under what conditions on σ_1^2 and σ_2^2 is $\mathbf{C}_X$ a valid covariance matrix?
- Suppose $\mathbf{Y} = [Y_1 \ Y_2]' = \mathbf{A}\mathbf{X}$ where $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$. For what values (if any) of σ_1^2 and σ_2^2 are the components Y_1 and Y_2 independent?

8.5.7 ■ The Gaussian random vector $\mathbf{X} = [X_1 \ X_2]'$ has expected value $E[\mathbf{X}] = \mathbf{0}$ and covariance matrix $\mathbf{C}_X = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$.

- Find the PDF of $W = X_1 + 2X_2$.

- (b) Find the PDF $f_{\mathbf{Y}}(\mathbf{y})$ of $\mathbf{Y} = \mathbf{A}\mathbf{X}$ where $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$.

8.5.8 ■ Let $\mathbf{X}$ be a Gaussian random vector with expected value $[\mu_1 \ \mu_2]'$ and covariance matrix

$$\mathbf{C}_{\mathbf{X}} = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}.$$

Show that $\mathbf{X}$ has bivariate Gaussian PDF $f_{\mathbf{X}}(\mathbf{x}) = f_{X_1, X_2}(x_1, x_2)$ given by Definition 5.10.

8.5.9 ■ $X = [X_1 \ X_2]'$ is a Gaussian random vector with $E[\mathbf{X}] = [0 \ 0]'$ and covariance matrix $\mathbf{C}_{\mathbf{X}} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.

- What conditions must a , b , c , and d satisfy?
- Under what conditions (in addition to those in part (a)) are X_1 and X_2 independent?
- Under what conditions (in addition to those in part (a)) are X_1 and X_2 identical?

8.5.10 ■ Let $\mathbf{X}$ be a Gaussian $(\mu_{\mathbf{X}}, \mathbf{C}_{\mathbf{X}})$ random vector. Let $\mathbf{Y} = \mathbf{A}\mathbf{X}$ where $\mathbf{A}$ is an $m \times n$ matrix of rank m . By Theorem 8.11, $\mathbf{Y}$ is a Gaussian random vector. Is

$$\mathbf{W} = \begin{bmatrix} \mathbf{X} \\ \mathbf{Y} \end{bmatrix}$$

a Gaussian random vector?

8.5.11 ■ The 2×2 matrix

$$\mathbf{Q} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

is called a rotation matrix because $\mathbf{y} = \mathbf{Q}\mathbf{x}$ is the rotation of $\mathbf{x}$ by the angle θ . Suppose $\mathbf{X} = [X_1 \ X_2]'$ is a Gaussian $(\mathbf{0}, \mathbf{C}_{\mathbf{X}})$ vector where $\mathbf{C}_{\mathbf{X}} = \text{diag}[\sigma_1^2, \sigma_2^2]$ and $\sigma_2^2 \geq \sigma_1^2$. Let $\mathbf{Y} = \mathbf{Q}\mathbf{X}$.

- Find the covariance of Y_1 and Y_2 . Show that Y_1 and Y_2 are independent for all θ if $\sigma_1^2 = \sigma_2^2$.
- Suppose $\sigma_2^2 > \sigma_1^2$. For what values of θ are Y_1 and Y_2 independent?

8.5.12 ♦ $\mathbf{X} = [X_1 \ X_2]'$ is a Gaussian $(\mathbf{0}, \mathbf{C}_{\mathbf{X}})$ vector where

$$\mathbf{C}_{\mathbf{X}} = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}.$$

Thus, depending on the value of the correlation coefficient ρ , the joint PDF of X_1 and X_2 may resemble one of the graphs of Figure 5.6 with $X_1 = X$ and $X_2 = Y$. Show that $\mathbf{X} = \mathbf{Q}\mathbf{Y}$, where $\mathbf{Q}$ is the $\theta = 45^\circ$ rotation matrix (see Problem 8.5.11) and $\mathbf{Y}$ is a Gaussian $(\mathbf{0}, \mathbf{C}_{\mathbf{Y}})$ vector such that

$$\mathbf{C}_{\mathbf{Y}} = \begin{bmatrix} 1 + \rho & 0 \\ 0 & 1 - \rho \end{bmatrix}.$$

This result verifies, for $\rho \neq 0$, that the PDF of X_1 and X_2 shown in Figure 5.6 is the joint PDF of two independent Gaussian random variables (with variances $1 + \rho$ and $1 - \rho$) rotated by 45° .

8.5.13 ♦ An n -dimensional Gaussian vector $\mathbf{W}$ has a block diagonal covariance matrix

$$\mathbf{C}_{\mathbf{W}} = \begin{bmatrix} \mathbf{C}_{\mathbf{X}} & \mathbf{0} \\ \mathbf{0} & \mathbf{C}_{\mathbf{Y}} \end{bmatrix},$$

where $\mathbf{C}_{\mathbf{X}}$ is $m \times m$, $\mathbf{C}_{\mathbf{Y}}$ is $(n-m) \times (n-m)$. Show that $\mathbf{W}$ can be written in terms of component vectors $\mathbf{X}$ and $\mathbf{Y}$ in the form

$$\mathbf{W} = \begin{bmatrix} \mathbf{X} \\ \mathbf{Y} \end{bmatrix},$$

such that $\mathbf{X}$ and $\mathbf{Y}$ are independent Gaussian random vectors.

8.5.14 ♦♦ In this problem, we extend the proof of Theorem 8.11 to the case when $\mathbf{A}$ is $m \times n$ with $m < n$. For this proof, we assume $\mathbf{X}$ is an n -dimensional Gaussian vector and that we have proved Theorem 8.11 for the case $m = n$. Since the case $m = n$ is sufficient to prove that $\mathbf{Y} = \mathbf{X} + \mathbf{b}$ is Gaussian, it is sufficient to show for $m < n$ that $\mathbf{Y} = \mathbf{A}\mathbf{X}$ is Gaussian in the case when $\mu_{\mathbf{X}} = \mathbf{0}$.

- Prove there exists an $(n-m) \times n$ matrix $\tilde{\mathbf{A}}$ of rank $n-m$ with the property that $\tilde{\mathbf{A}}\mathbf{A}' = \mathbf{0}$. Hint: Review the Gram-Schmidt procedure.

- (b) Let $\hat{\mathbf{A}} = \tilde{\mathbf{A}}\mathbf{C}_{\mathbf{X}}^{-1}$ and define the random vector

$$\bar{\mathbf{Y}} = \begin{bmatrix} \mathbf{Y} \\ \hat{\mathbf{Y}} \end{bmatrix} = \begin{bmatrix} \mathbf{A} \\ \hat{\mathbf{A}} \end{bmatrix} \mathbf{X}.$$

Use Theorem 8.11 for the case $m = n$ to argue that $\bar{\mathbf{Y}}$ is a Gaussian random vector.

- (c) Find the covariance matrix $\bar{\mathbf{C}}$ of $\bar{\mathbf{Y}}$. Use the result of Problem 8.5.13 to show that $\mathbf{Y}$ and $\hat{\mathbf{Y}}$ are independent Gaussian random vectors.

8.6.1 ● Consider the vector $\mathbf{X}$ in Problem 8.5.1 and define $Y = (X_1 + X_2 + X_3)/3$. What is the probability that $Y > 4$?

8.6.2 ■ A better model for the sailboat race of Problem 5.10.8 accounts for the fact that all boats are subject to the same randomness of wind and tide. Suppose in the race of ten sailboats, the finishing times X_i are identical Gaussian random variables with expected value 35 minutes and standard deviation 5 minutes. However, for every pair of boats i and j , the finish times X_i and X_j have correlation coefficient $\rho = 0.8$.

- (a) What is the covariance matrix of $\mathbf{X} = [X_1 \ \cdots \ X_{10}]'$?
- (b) Let

$$Y = \frac{X_1 + X_2 + \cdots + X_{10}}{10}.$$

What are the expected value and variance of Y ? What is $P[Y \leq 25]$?

8.6.3 ■ For the vector of daily temperatures $[T_1 \ \cdots \ T_{31}]'$ and average temperature Y modeled in Quiz 8.6, we wish to estimate the probability of the event

$$A = \left\{ Y \leq 82, \min_i T_i \geq 72 \right\}.$$

To form an estimate of A , generate 10,000 independent samples of the vector $\mathbf{T}$ and calculate the relative frequency of A in those trials.

8.6.4 ♦ We continue Problem 8.6.2 where the vector $\mathbf{X}$ of finish times has correlated components. Let W denote the finish time of the winning boat. We wish to estimate $P[W \leq 25]$, the probability that the winning boat finishes in under 25 minutes. To do this, simulate $m = 10,000$ races by generating m samples of the vector $\mathbf{X}$ of finish times. Let $Y_j = 1$ if the winning time in race j is under 25 minutes; otherwise, $Y_j = 0$. Calculate the estimate

$$P[W \leq 25] \approx \frac{1}{m} \sum_{j=1}^m Y_j.$$

8.6.5 ♦♦ Write a MATLAB program that simulates m runs of the weekly lottery of Problem 7.5.9. For $m = 1000$ sample runs, form a histogram for the jackpot J .